

SRIJANA KC KHATRI

Data Engineer

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California, USA

PROFESSIONAL SUMMARY

Results-driven **Data Engineer** with a strong background in designing and optimizing scalable data architectures for **Big Data, real-time streaming, and cloud-based solutions**. Experienced in building robust **ETL pipelines** using **Apache Spark, AWS Glue, and Azure Data Factory**, ensuring efficient data integration and transformation. Expertise in handling **structured and unstructured data** across **Hadoop, Snowflake, Redshift, and NoSQL databases** to drive insightful analytics. Proven ability to implement **real-time data ingestion and streaming solutions** with **Apache Kafka, Spark Streaming, and Flink**, enhancing data processing efficiency. Skilled in **SQL performance tuning, data warehousing, and machine learning integration**, enabling advanced analytics and predictive modeling. Adept at working with **cross-functional teams** to develop scalable, cloud-native architectures on **AWS and Azure**, improving data accessibility and decision-making. Passionate about **leveraging DevOps, CI/CD, and Agile methodologies** to automate workflows and optimize data-driven solutions.

EXPERIENCE

Data Engineer | United HealthCare | California, USA

Mar '22 - Present

- Designed and maintained scalable ETL pipelines using **Apache Spark** and **Hadoop** to process large-scale healthcare datasets, optimizing data workflows and speeding up data analysis.
- Implemented **Hadoop HDFS** for efficient data storage and retrieval, streamlining complex data operations and cutting down processing times significantly for healthcare data.
- Utilized **AWS Glue** for automating data integration from various sources, transforming raw healthcare data into useful insights that supported business intelligence and reporting needs.
- Enhanced data processing efficiency by applying **Hadoop MapReduce** and **Spark SQL** for distributed computing, drastically improving performance and speeding up data processing for large datasets.
- Developed **data warehousing solutions** with **Amazon Redshift**, storing large volumes of data and boosting query performance by 30%, enabling faster analysis and reporting for healthcare operations.
- Built real-time data streaming pipelines using **Apache Kafka** and **Spark Streaming**, enabling the healthcare team to analyze and act on operational data as it was being generated.
- Created and managed interactive **Tableau dashboards** to visualize key healthcare metrics, providing actionable insights for stakeholders and enhancing decision-making across the organization.
- Developed and optimized data pipelines for both **batch processing with Hadoop** and **real-time streaming with Apache Kafka and Spark**, ensuring the timely processing of large datasets across different use cases.
- Leveraged **Python** and **SQL** for data transformation and automation, streamlining manual workflows and ensuring that data was accurate, timely, and ready for analysis.
- Implemented **robust data governance frameworks** to ensure data integrity, compliance with **HIPAA regulations**, and adherence to security best practices for handling sensitive patient information.
- Collaborated closely with **cross-functional teams** to define data requirements, improve reporting workflows, and ensure analytics solutions met organizational goals and aligned with market trends.
- Applied **machine learning algorithms** to healthcare data, identifying trends, predicting patient outcomes, and providing operational insights to improve healthcare service delivery.

Environment: *Apache Spark, Hadoop, Apache Kafka, Spark Streaming, AWS Glue, Amazon Redshift, HDFS, Hadoop MapReduce, Hive, SQL, Python, Tableau, NoSQL, Machine Learning, HIPAA Compliance.*

Data Engineer | Cedar Gate | Kathmandu, Nepal

Oct '20 – Nov '21

- Designed and developed **ETL pipelines** using **Azure Data Factory**, automating the process of extracting, transforming, and loading data from various sources into a centralized storage system. This streamlined workflows and improved data access.
- Used **Databricks** and **Scala** for processing large datasets with **Apache Spark**, enabling advanced data transformations and allowing integration of **machine learning models** for predictive insights and better decision-making.
- Built **data models** and optimized **storage solutions** in **Azure Synapse Analytics**, making data aggregation and reporting much more efficient and significantly improving query performance.
- Implemented **real-time data processing** using **NoSQL databases** like **Cassandra** and **MongoDB**, ensuring high availability and scalability while handling unstructured data, which made the system more flexible.
- Created a fault-tolerant data processing system using **Apache Spark** and **Azure Databricks**, ensuring faster and more reliable transformations for large-scale healthcare data.

- Developed interactive **Power BI dashboards**, providing stakeholders with clear, actionable insights into key business metrics, which helped improve operational decision-making.
- Streamlined **data storage** and retrieval using **Azure Data Lake** and **Blob Storage**, allowing us to handle and process large amounts of healthcare data with greater speed and efficiency.
- Collaborated with cross-functional teams to design and integrate **machine learning models** within the data pipeline, which enhanced the accuracy of predictive analytics and provided deeper insights into healthcare operations.

Environment: Azure Data Factory, Databricks, Apache Spark, Power BI, NoSQL (Cassandra, MongoDB), Scala, Azure Synapse Analytics, Azure Data Lake, Cloud Data Solutions, Machine Learning Integration, ETL Pipelines, Real-Time Data Processing.

Data Engineer | Verisk | Kathmandu, Nepal

Oct '18 – Sep '20

- Developed and managed a **Hadoop-based data lake** to consolidate vehicle telemetry data, enabling scalable data storage and efficient retrieval for advanced analytics.
- Designed **real-time data ingestion pipelines** using **Apache Kafka, Hadoop, and Spark Streaming**, reducing latency in insurance risk modeling and claims processing.
- Implemented **distributed computing solutions** using **Apache Spark**, improving batch and streaming data processing efficiency for large-scale datasets.
- Optimized **data warehousing solutions** in **Snowflake**, ensuring faster query execution and cost-effective storage management for insurance analytics.
- Automated **ETL processes** using **Apache Airflow, AWS Glue, and SQL**, streamlining data transformations and reducing manual effort in data pipeline management.
- Leveraged **Apache Hive and Pig for big data processing**, enabling seamless data transformation and advanced aggregations for structured and unstructured datasets.
- Integrated **NoSQL databases** for managing semi-structured and unstructured data, enhancing data flexibility in fraud detection and telematics analytics.
- Built **microservices and API-driven architectures** using **Java and Python**, enabling real-time data access for underwriting, fraud analysis, and market trend forecasting.
- Implemented **DevOps best practices** with **Docker, Kubernetes, and CI/CD pipelines**, ensuring scalable and efficient deployment of data pipelines.
- Collaborated with **data scientists and business teams** to refine **predictive risk models**, improving accuracy in fraud detection and dynamic pricing strategies.
- Applied **machine learning and advanced analytics** to insurance data, identifying hidden patterns in vehicle telemetry and claims data for risk assessments.
- Developed **real-time dashboards in Tableau and Power BI**, visualizing key metrics and insights for insurance policy adjustments and underwriting strategies.

Environment: Apache Spark, Hadoop, Apache Hive, Apache Pig, Apache Kafka, Spark Streaming, Flink, Snowflake, SQL, Apache Airflow, AWS Glue, Java, Python, Scala, Shell Scripting, AWS (S3, Lambda, EC2), Docker, Kubernetes, Tableau, Power BI.

TECHNICAL SKILLS

Data Engineering Tools:	Apache Spark, MapReduce, Hive, Databricks, Data Modeling, Snowflake, Data warehouse, Teradata, Hadoop distributed system, Kafka, CI/CD Pipeline, Apache Airflow, Apache Kafka, Spark Streaming
ETL Tools:	Talend, Informatica, Apache Airflow.
Database platforms:	Teradata, HBase, Cassandra, Oracle Database, MySQL, MongoDB, SSIS, SQL Server Reporting Services (SSRS), PostgreSQL.
Microsoft Azure:	Azure Data Factory, Azure VM, Azure Databricks, Azure Data Lake Storage, Azure Event Hubs, Azure Synapse Analytics, Azure HDInsight, Azure Blob Storage.
AWS:	AWS, S3, EC2, Lambda, Glue, AWS Lake Formation, Kinesis Data Firehose, EMR
Programming Languages:	Python, R, Structured query language (SQL), PySpark, Scala, Java, Spark-SQL, UNIX
Python lib:	Pandas, SciPy, Scikit-learn, Seaborn, TensorFlow, Beautiful Soup, NumPy
Data Visualization:	Power BI, Tableau, Excel, Matplotlib, Looker, Excel.
GCP:	Big Query, Google storage.
Software & Model Deployment:	Git, Agile methodologies, Jira, Scrum, Docker, Kubernetes.

EDUCATION

University of Cumberland | Master's degree | **Master of Science (MS) | Information Technology**